Synopsis Presentation

on

NetAgent: AI-Powered Network Security Sentinel

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1. Introduction

1.1 Overview

- The modern industrial world uses vast arrays of connected IoT devices, leading to decentralized computer networks and a massively expanded attack surface.
- Network administrators face a **"Cognitive Gap"**—they are overwhelmed by thousands of minor warning messages from traditional firewalls.
- Due to this noise, sophisticated, slow-moving threats (like covert SQL injections or data exfiltration) often bypass human detection, causing severe damage.

1.2 Purpose

- Current defenses are reactive and static. Nmap or Wireshark provide raw data but demand heavy manual expertise and time.
- **NetAgent** is conceived as an autonomous **"SOC-in-a-Box"**. It actively reasons through complex, multi-layered security data.
- It acts as a "virtual Tier-3 analyst," correlating data to provide immediate, contextually rich **Actionable Intelligence** (What happened, Why, and How to stop it).

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2. Literature Review

- Summary of existing solutions and their limitations in current network security environments.

Sr. No.	Name of Solution/System	Features	Limitations/ Drawbacks
1.	**Nmap** (Gordon Lyon)	Active scanning, OS fingerprinting, robust scripting engine. Maps network topology.	Provides raw data only. Lacks context and reasoning; requires manual analysis.
2.	**Scapy** (Philippe Biondi)	Deep packet inspection (DPI), custom packet crafting and protocol manipulation.	Highly manual tool requiring deep protocol expertise. Not autonomous.
3.	**Traditional IDS** (Snort, Suricata)	Detects known threats using rigid, pre-defined attack signatures.	Fundamentally reactive. Blind to zero-day exploits and polymorphic malware

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3. Problem Statement

The Cognitive Gap & Log Fatigue

- **Unstructured Data Overload:** Existing security stacks fail to autonomously convert high-volume raw telemetry into structured context, overwhelming analysts.
- **Critical Delays:** The Mean Time to Action (MTTA) typically spans **2 to 4 hours** as human analysts manually sift through and correlate logs from disparate sources.
- **Expanded Vulnerability Window:** This protracted delay allows sophisticated threats to persist and escalate before a response is mounted.

Objective: Reduce the MTTA from hours to under 10 minutes by synthesizing a high-fidelity report detailing WHAT happened, WHY it succeeded, and HOW to respond

4. Proposed Solution

NetAgent utilizes a **Hybrid Agentic Architecture** powered by the ReAct (Reasoning and Acting) framework, physically separating sensing from AI reasoning.

1. The Probe (Raspberry Pi)

Acts as the frontline sensor performing targeted Deep Packet Inspection (Scapy) and active reconnaissance (Nmap/Nikto). It curates essential metadata to eliminate bulk log data.

2. The Brain (Edge Compute)

Powered by a locally quantized LLM (Llama 3). Uses LangChain to recursively orchestrate tools, self-correct, and analyze the curated metadata entirely offline.

3. Recursive Validation Loop

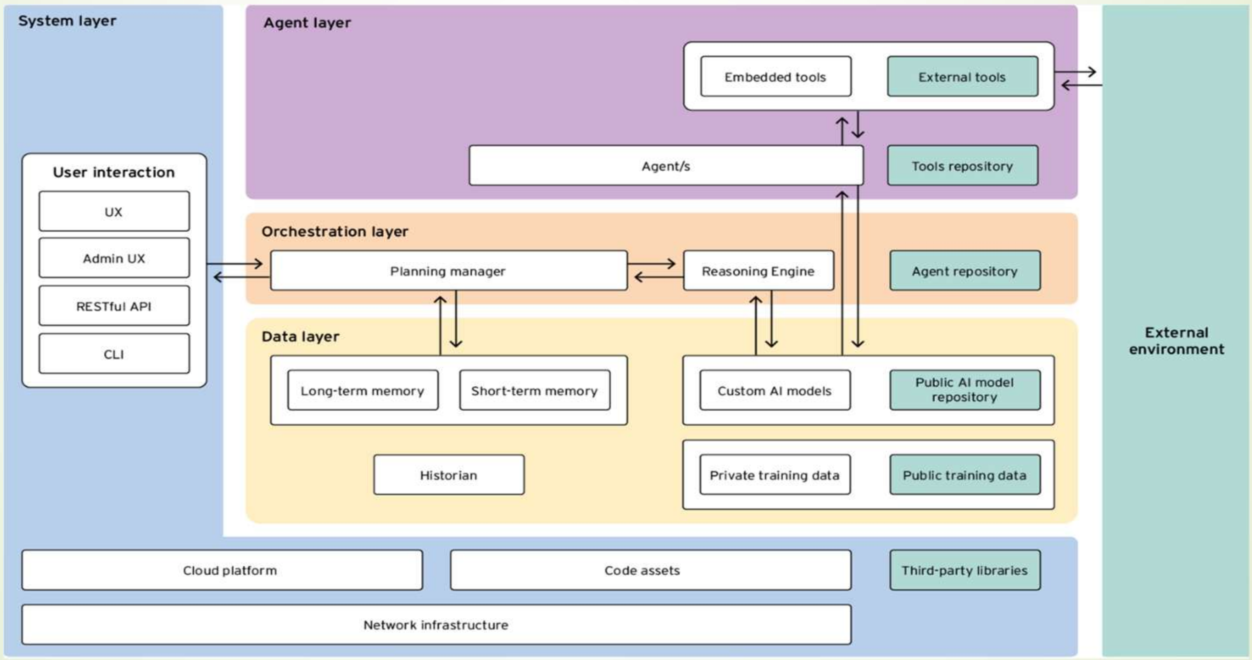
The AI acts as a master orchestrator. It executes specialized tools, analyzes the output, and initiates the next reasoning loop. This continuous, self-correcting investigation minimizes false positives and generates 100% private, human-readable Actionable Intelligence.

5. Objectives

- **Automate Orchestration** Achieve 100% autonomous execution of multi-stage security tools (Nmap, Nikto, Searchsploit) via an LLM, eliminating human intervention in the initial analysis.
- **Reduce Latency** Drastically reduce MTTA by generating a comprehensive threat analysis and a detailed, step-by-step remediation guide in under 10 minutes.
- **Ensure Data Sovereignty** Guarantee that all AI reasoning, machine learning inference, and packet inspection are performed 100% locally on edge hardware to comply with privacy regulations.
- **Eliminate Log Fatigue** Condense thousands of raw alerts into a single, structured report articulating the "What, Why, and How" of high-risk findings.

6. Theoretical Analysis

6.1 Block Diagram



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6. Theoretical Analysis

Type	Item	Description
Hardware	Raspberry Pi 4/5	Frontline sensor acting as the network probe for DPI and active scanning.
Hardware	Edge Node (Laptop/PC)	Hosting the LLM (Ollama) to perform heavy reasoning preventing Pi memory crash.
Software	Python 3.10+	Primary language for tool integration and agent orchestration logic.
Software	Ollama / Llama 3	Local, quantized LLM engine for highly capable offline reasoning and analysis.
Software	LangChain / CrewAI	Agentic framework for managing the ReAct "Chain of Thought" loop.
Software	Security Stack	Nmap, Nikto, Scapy, and Searchsploit binaries manipulated by the AI.

Applications

NetAgent functions as a Portable, Autonomous Vulnerability Auditor (PAVA).

- **SMB Internal Audits** A plug-and-play solution providing on-demand, non-intrusive security snapshots for small businesses lacking dedicated cybersecurity teams.
- **Branch Verification** Can be shipped to remote offices to autonomously assess local network segments and verify compliance with corporate security policies.
- **Change-Management** Quickly validates network security integrity immediately after new hardware, servers, or critical firewall configurations are deployed.
- **Educational Tool** Serves as a fully contained, low-cost platform in cyber-range environments for students learning network reconnaissance and agentic AI.

GitHub Link

- **NetAgent Source Code Repository**

https://github.com/Sarthak-Purohit/NetAgent

REFERENCES

- **Lyon, G. (1997):** Nmap: The Network Mapper. Foundational study on TCP/IP service fingerprinting and active network reconnaissance.
- **Biondi, P. (2003):** Scapy and the Art of Packet Crafting. Core source for Deep Packet Inspection (DPI) and dynamic protocol manipulation.
- **Touvron, H., et al. (2023):** Llama: Open Foundation Language Models. Foundational research enabling high-performance local LLM execution.
- **Chase, H. (2022):** LangChain Documentation. Methodology for LLM tool-calling, memory management, and ReAct orchestration.
- **Offensive Security:** The Exploit Database (Searchsploit). Primary source for vulnerability correlation and contextualization data.

Thank You

Queries ?